SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA02 **COURSE NAME:C Programming**

COURSE OUTCOME COVERED

CO4: *Implement structures, unions, and nested structures for modeling and processing real-world data.* (BL4,BL5)

Assessment Tool Weightage: 35 %

QUESTIONS **Total Marks: 100**

CASE STUDY

Code of Conduct:

I K.Kavya 192425183 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

84/100

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL2 ~ CASE STUDY

16. POWER DISTRIBUTION MONITORING SYSTEM

1. Structure design

```
struct address
{
  char city [30];
   char state [30];
    char pincode [10];
     };
```

department structure

```
struct department
{ int deptid;
   char dept_name [30];
    };
```

Nested Employee structure

```
struct Employee
{ int id;
  char name [30];
  struct address addr;
  struct Department dept;
   };
```

2. union for role information

since an employee can have only one at role at a time, a union saves memory

```
union role
{
char administrator [20];
char operator [20];
```

```
char technician[20];
    };
```

Benifit

only one member occupies memory at atime

Eample

```
union role r;
Stracpy (r.administrator, "Administrator");
```

memory required = size of largest memory only

3. union for employment category

only one employment category is applicable

```
union Employement type
  {char fulltime [15];
   char contract [15];
   };
```

Example

```
union Employementtype emp;
 strcpy (emp, fulltime, "Fulltime");
```

4. Union for operational status

only one status exists at atime

```
union status
 {
 char active [10];
 char inactive[10];
 char maintaince [15]
   };
```

Example

```
union status s;
```

strcpy (s.active, "Active");

5- storage classes

static variable

used to retain value throughout prog execution

static int totalrecord = 0;

advantages

- value persists between function calls
- useful for counting record

Extern variable

used for sharing data among multiple source files:

int systemRecords = 100;

extern int systemrecords;

advantages:

- global access
- avoids duplicate copies of data

6 complete Data model

```
struct Employee
{
    int id;
    char name[30];
    struct address addr;
    struct Department dept;
    union Role role;
    union EmploymentType emptype;
    union status sttatus
};
```

Analysis

Nested structures

- Organize related information logically
- Improve readability and maintainance
- represent real-world entities effectively

union

- Store only one value at a time
- Reduce memory consumption
- suitable for role, employment type and occurs

Static storage class

- maintains data type throughout program executions.
- useful for counters and persistent records

Extern storage class

- allow centralized access across modules
- Enables data sharing among files

conclusion:

the power distribution monitoring systems can be efficiently modeled using nesting structures for employee, address and department details. unions minimize memory usage by storing only one role, employment category and operational status at a time, static variables preserve data during executi-on, while extern variables provides centralized access across program modules, improving memory management and persistence

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20%	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10%	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand